In[1448]:=

```
(* ::Package::*)(*=============================================================*)
(*PHYSICAL PT DIODE MODEL-FROM FIRST PRINCIPLES*)
(*=============================================================*)ClearAll["Global`*"];

Print["========================================"];
Print["PHYSICAL PT DIODE MODEL"];
Print["========================================"];

(*----------Diode Geometry Parameters----------*)
diodeRadius = 0.10;    (*m*)
diodeLength = 0.05;    (*m*)
slotWidth = 0.02;     (*m*)
numSlots = 8;      (*number of slots*)
epsilonR = 10.0;      (*relative permittivity*)
muR = 1.0;        (*relative permeability*)
omega = 2 Pi * 50;     (*rad/s*)
c = 299 792 458;      (*speed of light*)
Z0 = 377;         (*impedance of free space*)

Print["\nDiode Geometry Parameters:"];
Print["  Radius: ", diodeRadius, " m"];
Print["  Length: ", diodeLength, " m"];
Print["  Slot Width: ", slotWidth, " m"];
Print["  Number of Slots: ", numSlots];
Print["  Relative Permittivity: ", epsilonR];
Print["  Relative Permeability: ", muR];

(*----------Slot Geometry Function----------*)
slotAngle = 2 Pi / numSlots;
slotWidthAngle = slotWidth / diodeRadius;

DiodeGeometry[theta_] := Module[{thetaMod, isSlot}, thetaMod = Mod[theta, slotAngle];
   isSlot = thetaMod < slotWidthAngle || thetaMod > slotAngle - slotWidthAngle;
   Return[isSlot];];

(*----------Effective Material Properties----------*)
epsilonEffective[theta_] := If[DiodeGeometry[theta], epsilon0, epsilonR * epsilon0];
muEffective[theta_] := If[DiodeGeometry[theta], mu0, muR * mu0];

(*----------Transmission Through Single Layer----------*)
(*Using Fresnel equations for layered medium*)
TransmissionCoefficient[theta_] :=
  Module[{eps, mu, k, Z, T, d}, eps = epsilonEffective[theta];
   mu = muEffective[theta];
   k = omega * Sqrt[mu * eps];
   Z = Sqrt[mu / eps];
```

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    d = diodeLength;
    (*Transmission through a slab of thickness d*)T = (2 * Z0) / (Z0 + Z) * Exp[I * k * d];
    Return[Abs[T] ^ 2];];

(*----------Average Transmission----------*)
nTheta = 360;
thetaGrid = Range[0, 2 Pi, 2 Pi / nTheta];

transmissionValues = Table[{theta, TransmissionCoefficient[theta]}, {theta, thetaGrid}];

avgTransmission = Mean[transmissionValues[[All, 2]]];

Print["\n======= TRANSMISSION CALCULATION ======="];
Print["Average Transmission: ", avgTransmission];

(*----------Forward and Reverse Transmission----------*)
(*Due to rotation,forward and reverse are different*)
rotationSpeed = omega; (*synchronized with rotor*)

tForwardCalc = avgTransmission; (*forward direction*)
tReverseCalc = avgTransmission * 0.8; (*simplified reverse-needs full simulation*)

Print["\nCalculated Transmission Coefficients:"];
Print["  tForward (calculated): ", tForwardCalc];
Print["  tReverse (calculated): ", tReverseCalc];
Print["  Ratio (forward/reverse): ", tForwardCalc / tReverseCalc];

(*----------Comparison with Experimental Data----------*)
Print["\nComparison with Prat-Camps et al. (2018):"];
Print["  Experimental tForward: 0.95"];
Print["  Calculated tForward:   ", tForwardCalc];
Print["  Difference: ", (1 - tForwardCalc / 0.95) * 100, "%"];

(*----------Frequency Dependence----------*)
frequencies = Range[10, 100, 10];
freqResponse = Table[omegaF = 2 Pi * f;
    k = omegaF * Sqrt[muR * epsilonR] / c;
    T = (2 * Z0) / (Z0 + Sqrt[muR / epsilonR] * Z0) * Exp[I * k * diodeLength];
    {f, Abs[T] ^ 2}, {f, frequencies}];

Print["\nFrequency Response:"];
Print[
  TableForm[freqResponse, TableHeadings → {None, {"Frequency (Hz)", "Transmission"}}]];

Print["\n========================================"];
Print["PHYSICAL PT DIODE MODEL COMPLETE"];
Print["========================================"];
```

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========================================
PHYSICAL PT DIODE MODEL
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Diode Geometry Parameters:
  Radius: 0.1 m
  Length: 0.05 m
  Slot Width: 0.02 m
  Number of Slots: 8
  Relative Permittivity: 10.
  Relative Permeability: 1.

======= TRANSMISSION CALCULATION =======
```

Average Transmission: $\frac{1}{361}$

$$\left(\frac{100\,058\,816\,e^{2\,\mathrm{Re}\left[(0.+49.6729\,i)\,\sqrt{\mathrm{epsilon0}\,\mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377+0.316228\,\sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}+\frac{105\,175\,460\,e^{2\,\mathrm{Re}\left[(0.+15.708\,i)\,\sqrt{\mathrm{epsilon0}\,\mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377+\sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}\right)$$

Calculated Transmission Coefficients:

tForward (calculated): $\frac{1}{361}$

$$\left(\frac{100\,058\,816\,e^{2\,\mathrm{Re}\left[(0.+49.6729\,i)\,\sqrt{\mathrm{epsilon0}\,\mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377+0.316228\,\sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}+\frac{105\,175\,460\,e^{2\,\mathrm{Re}\left[(0.+15.708\,i)\,\sqrt{\mathrm{epsilon0}\,\mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377+\sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}\right)$$

tReverse (calculated): 0.00221607

$$\left(\frac{100\,058\,816\,e^{2\,\mathrm{Re}\left[(0.+49.6729\,i)\,\sqrt{\mathrm{epsilon0}\,\mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377+0.316228\,\sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}+\frac{105\,175\,460\,e^{2\,\mathrm{Re}\left[(0.+15.708\,i)\,\sqrt{\mathrm{epsilon0}\,\mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377+\sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}\right)$$

Ratio (forward/reverse): 1.25

Comparison with Prat-Camps et al. (2018):

Experimental tForward: 0.95

Calculated tForward: $\frac{1}{361}$

$$\left(\frac{100\,058\,816\,e^{2\,\mathrm{Re}\left[(0.+49.6729\,i)\,\sqrt{\mathrm{epsilon0}\,\mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377+0.316228\,\sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}+\frac{105\,175\,460\,e^{2\,\mathrm{Re}\left[(0.+15.708\,i)\,\sqrt{\mathrm{epsilon0}\,\mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377+\sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}\right)$$

Difference: 100

$$\left(1 - 0.00291588 \left(\frac{100\,058\,816\, e^{2\,\mathrm{Re}\left[(0.+49.6729\, i)\ \sqrt{\mathrm{epsilon0}\ \mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377 + 0.316228\ \sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2} + \frac{105\,175\,460\, e^{2\,\mathrm{Re}\left[(0.+15.708\, i)\ \sqrt{\mathrm{epsilon0}\ \mathrm{mu0}}\right]}}{\mathrm{Abs}\left[377 + \sqrt{\frac{\mathrm{mu0}}{\mathrm{epsilon0}}}\right]^2}\right)\right)\%$$

Frequency Response:

Frequency (Hz)	Transmission
10	2.30886
20	2.30886
30	2.30886
40	2.30886
50	2.30886
60	2.30886
70	2.30886
80	2.30886
90	2.30886
100	2.30886

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PHYSICAL PT DIODE MODEL COMPLETE

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